Exercise list — Topic 5: Nonverifiable income, strategic default, termination, debt structure

Answer key (full derivations in `lista-topic5-nonverifiable-income-termination-solutions.pdf`)

1. Control passing to creditors

Equity $= \max(V - F, 0)$ is convex (a call), debt $= \min(V, F)$ is concave. In control, shareholders gamble (asset substitution), underinvest (debt overhang), favour quick payoffs and resist liquidation; after bankruptcy, creditors cut risk, sell assets and liquidate — sometimes too early, since they ignore the upside above F. Eastern Airlines ($F = 500$; continue 600/200, liquidate 480): equity 50 vs. 0 (continue), debt 350 vs. 480 (liquidate), firm 400 vs. 480. Own example ($F = V_L = 500$; continue 800/300, worth 550): creditors get $400 < 500$ and liquidate. Conversion of debt into equity in Chapter 11 realigns incentives with total value.

2. Proving (5.11)

$P_u = f(1-e)$ (only good, non-downgraded projects are upgraded), $V_s^u = r_1$; break-even $f(1-e)r_1 + [1-(1-e)f]V_s^d = R$ gives

$$r_1 = \frac{R - [1-(1-e)f]V_s^d}{(1-e)f}.$$

Fixed point: $r_1 = R$ if $R \le \max(L, q^d X/R)$; otherwise $V_s^d = \max(L, q^d X/R) < R < r_1$. Matches the book (p. 268).

3. $\beta_\circ$ decreases in q

$$\frac{\partial \beta_\circ}{\partial q} = \frac{-K\left(\frac{y}{2} - \frac{\alpha^2 y^2}{4\bar{c}}\right)}{\left[q\frac{y}{2} + (1-q)\frac{\alpha^2 y^2}{4\bar{c}}\right]^2} \le 0,$$

since $\frac{y}{2} - \frac{\alpha^2 y^2}{4\bar{c}} = \frac{y}{2}\left[1 - \alpha\frac{\alpha y}{2\bar{c}}\right] \ge 0$ by $\alpha \le 1$ and $\alpha y/2 \le \bar{c}$ (zero only if $\alpha = 1$, $y/2 = \bar{c}$). The strategic-default recovery $y/2$ exceeds the liquidity-default recovery $L_\circ$. Matches the book (p. 267).

4. Renegotiation under entrepreneur control

Without renegotiation: $\pi_{wt}^E = q l_g^g + (1-q) l_g^b$. In θ_b the entrepreneur sells the switch to a_b for $T = y_b^b - y_g^b$, so $\pi_w^E = q l_g^g + (1-q)(l_b^b + y_b^b - y_g^b)$ and

$$\pi_w^E - \pi_{wt}^E = (1-q)\left[(y_b^b + l_b^b) - (y_g^b + l_g^b)\right] > 0,$$

while the investor stays at π_2. Matches the book (p. 266), whose middle lines have sign typos.

5. Diamond (1991), no liquidation

$q = \frac{3}{4}$, $q^d = \frac{2}{3}$. (a) $\rho = R^2/q = \frac{4}{3}$; $r_1 = 1$ (riskless, $V_s^d = r_1$, since $q^d X/R = \frac{7}{6} \ge 1$). (b) Long-term: nothing due at $t = 1$; short-term: rollover at face 1 (upgrade) or $\frac{3}{2} \le X$ (downgrade), lenders break even ($\frac{2}{3} \cdot \frac{3}{2} = 1$): no default, no liquidation. (c) $U_l = X - \rho = \frac{5}{12}$. (d) $e = f^d(1-f)/[f(1-f^d)] = \frac{1}{2}$. (e) $U_s = \frac{1}{2}(\frac{7}{4} - \frac{3}{2}) + \frac{1}{2}(\frac{7}{4} - 1) = \frac{1}{2} > \frac{5}{12}$: short-term debt preferred ($V_s^d = 1 > V_l^d = \frac{8}{9}$, Lemma 5.2.1). No worked answer in the book (cites Diamond 1991).

6. Bolton–Scharfstein with general contracts

(PD) unchanged: $D \geq (z^S - z^F)(M - D)$. (IC′): $(\Delta p)[z^S R_b^{SS} + (1 - z^S) r_b^S - z^F R_b^{FS}] \geq B_0$. (IR′): $D + z^S(D - I - R_b^{SS}) - (1 - z^S) r_b^S \geq I - A$. r_b^S and R_b^{SS} enter only via $W^S = z^S R_b^{SS} + (1 - z^S) r_b^S$, so shift all of it into R_b^{SS}: $r_b^S = 0$. R_b^{FS} only hurts (IC′), so $R_b^{FS} = B/\Delta p \leq R_b^{SS}$. With (PD) binding: $z^S(R_b^{SS} - R_b^{FS}) + \frac{D}{M-D} R_b^{FS} = \frac{B_0}{\Delta p}$. Matches the book (pp. 603–604).

7. Liquidation with a private signal

(i) γ is a sufficient statistic: pay $B/\Delta p$ after $\bar{\gamma}$, 0 after $\underline{\gamma}$; liquidate iff $\underline{\gamma}$; $\mathcal{P} = p_H \bar{\gamma} R + (1 - p_H) L - p_H B/\Delta p \geq I - A \iff A \geq \bar{A}$. (ii) Truth-telling $\bar{\gamma} R_b \geq L_b \geq \underline{\gamma} R_b$ and effort $(\Delta p)(\bar{\gamma} R_b - L_b) \geq B$ bind at $R_b = B/[(\Delta p)(\Delta\gamma)]$, $L_b = \underline{\gamma} R_b$; the rent rises by $\underline{\gamma} B/[(\Delta p)(\Delta\gamma)]$, so $A \geq \bar{A} + \underline{\gamma} B/[(\Delta p)(\Delta\gamma)]$. Matches the book (pp. 577–578).

8. Decreasing returns and debt maturity (Tirole 5.7)

(i) Maximise $U_b = rI + F(\rho^*) p_H R(I) - [1 + \int_0^{\rho^*} \rho f] I$ s.t. $rI + F(\rho^*) p_H [R(I) - BI/\Delta p] \geq I - A + [\int_0^{\rho^*} \rho f] I$ (binding). FOC in ρ^*: $\rho^* = \frac{\rho_1(I) + \mu \rho_0(I)}{1 + \mu}$, with $\rho_1 = p_H R(I)/I$, $\rho_0 = p_H[R(I)/I - B/\Delta p]$. Eliminating μ with the FOC in I:

$$p_H \Big[R'(I) - \frac{R(I)}{I} \Big] = \frac{1 - r - \int_0^{\rho^*} F(\rho)\, d\rho}{F(\rho^*)} \qquad (\text{CRS: } \textstyle\int_0^{\rho^*} F = 1 - r).$$

(ii) Left side decreasing in I (assumption); right side decreasing in ρ^* (needs $f[\int_0^{\rho^*} F - (1 - r)] < F^2$, true for the uniform), so I and ρ^* comove. In the binding breakeven, $\partial A/\partial I = F p_H B/[(1 + \mu)\Delta p] > 0$ and $\partial A/\partial \rho^* = I f(\rho^*)(\rho^* - \rho_0) > 0$, so lower $A \Rightarrow$ lower I and ρ^*. Short-term debt per unit, $r - \rho^*$, rises: weaker firms borrow shorter.

9. The priority game (Tirole 5.9)

(i) $\max_{I_1} \{r - I_1 + [p + \tau(I_1)] R\} \Rightarrow \tau'(I_1^*) R = 1$, $0 < I_1^* < r$. Any $(r_b, r_l, R_b, R_l) \geq 0$ with $r_b + r_l + I_1^* = r$, $R_b + R_l = R$, $I = r_l + [p + \tau(I_1^*)] R_l$ implements it: one degree of freedom (e.g. $r_l = r - I_1^*$, or $r_l = 0$).
(ii) $r - I_1^* < I$ forces $R_l > 0$, $R_b < R$. Secret deal $+\delta r_l$ for $\delta R_l = \delta r_l/[p + \tau(I_1)]$, $\delta I_1 = -\delta r_l$: $\delta U_b = [1 - \tau'(I_1) R_b] \delta r_l = (R_l/R) \delta r_l > 0$ at I_1^* — paid by the other lenders. Collusion-proofness: $\tau'(I_1) R_b = 1 \Rightarrow I_1 < I_1^*$; the entrepreneur maximises R_b, so $r_b = 0$, $r_l = r - I_1$, and $I = r - I_1 + [p + \tau(I_1)](R - R_b)$. Yes, the indeterminacy is resolved, with a short-term bias.

10. One vs. two creditors (Amaro de Matos 5.2.5 #2)

Shapley bargaining (manager/buyer, creditor A, creditor B; worths y, y^A, y^B, $\Delta = y - y^A - y^B > 0$): strategic default: $S = \frac{2y + y^A + y^B}{6} = \frac{y}{2} - \frac{\Delta}{6}$, so creditors get $y - S = \frac{y}{2} + \frac{\Delta}{6} > \frac{y}{2}$ (paid more). Liquidity default: buyer gets $\alpha(\frac{y}{2} - \frac{\Delta}{6})$, buys with probability $\frac{\alpha(3y - \Delta)}{6\bar{c}}$; creditors get $\frac{\alpha(3y + \Delta)}{6}$: $\overline{L}_\circ = \frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2 \Delta^2}{36\bar{c}} < L_\circ$ (paid less). The $\beta_\circ$ denominator is $q(\frac{y}{2} + \frac{\Delta}{6}) + (1 - q)\overline{L}_\circ$: the book's $q(\frac{y}{2} - \frac{\Delta}{6})$ (p. 138) is a sign error. Two creditors liquidate less iff $q > \alpha^2 \Delta/(6\bar{c} + \alpha^2 \Delta)$. Example $y = 10$, $\Delta = 3$, $\alpha = 0.8$, $\bar{c} = 6$, $q = 0.6$, $K = 2$: $y - S = 5.5$ vs 5; $\overline{L}_\circ = 2.64$ vs 2.666667; $\beta_\circ = 0.459137$ vs 0.491803.

11. Liquidating debt preferred (Amaro de Matos 5.2.5 #4)

Bayes: $P_d = 1 - (1 - e) f$, $f^d = fe/P_d$. Then $U^N = X + C - \frac{R^2}{f} + eRV^d(\frac{1}{f^d} - \frac{1}{q^d})$, $U^L = (1 - e)(X + C) - \frac{R^2}{f} + \frac{eRL}{f^d}$, so $U^L - U^N = e(\widehat{C} - C)$. Non-liquidating debt has $V^d \leq q^d X/R$, hence

$\widehat{C} \geq \frac{LR-q^dX}{f^d}$; and (5.12) $\iff C \leq \frac{LR-q^dX}{f^d}$. So (5.12) $\Rightarrow C \leq \widehat{C} \Rightarrow U^L \geq U^N$ (strict except at a knife-edge). Also $\widehat{C} \geq 0$ needs $LR \geq q^dX$ (credible liquidation).

12. Diamond (1991) examples (Amaro de Matos 5.2.5 #5–#6)

Part 0 (#5). $q = \frac{3}{4}$, $\rho = \frac{4}{3}$; $e = \frac{1}{2}$, $P_d = \frac{3}{4}$, $q^d = \frac{2}{3}$; $r_1 = 1$ (riskless). (b) $q^dX/R = \frac{7}{6} > r_1$ and $> L$: refinance at $F^d = \frac{3}{2} \leq X$, no liquidation. (c) $U_{LT} = \frac{7}{4} - \frac{4}{3} = \frac{5}{12}$. (d) $e = \frac{1}{2}$. (e) $U_{ST} = \frac{1}{2}(\frac{7}{4} - \frac{3}{2}) + \frac{1}{2}(\frac{7}{4} - 1) = \frac{1}{2} > \frac{5}{12}$: short-term preferred.
#6, $f = \frac{1}{2}$ ($X = 1.4$, $C = 0.35$): $q^dX/R = \frac{14}{15} = 0.933333 < L = 1 \Rightarrow$ downgraded short-term borrowers are liquidated; $r_1 = 1$. $U_{ST} = \frac{1}{2}(1.75-1) = 0.375 < U_{LT} = 1.75 - \frac{4}{3} = 0.416667$: long-term preferred ($\widehat{C} = 0.266667 < C$).
#6, $f = \frac{3}{4}$ ($f^d = \frac{1}{3}$ kept): $e = \frac{1}{6}$, $q = \frac{7}{8}$, $\rho = \frac{8}{7}$, still liquidation, $r_1 = 1$. $U_{ST} = \frac{5}{6}(0.75) = 0.625 > U_{LT} = 1.75 - \frac{8}{7} = 0.607143$: reversed ($\widehat{C} = 0.457143 > C$; Prop. 5.2.1 bound 0.6 lies between $\frac{1}{2}$ and $\frac{3}{4}$).

13. Borrower-friendly court (Tirole 4.9)

(i) r, L to investors (no incentive cost; raises pledgeable income); $R_b = B/\Delta p$ (any excess is a transfer that forces more liquidation). (1): continuation always efficient; (2): always liquidating finances the project, always continuing does not. **(ii)** $\max_{r^*} E[r]+\int_{r^*}^{\bar{r}} \rho_1 f + F(r^*)L - I$ s.t. $E[r]+\int_{r^*}^{\bar{r}} \rho_0 f + F(r^*)L \geq I - A$; NPV falls in r^*, so r^* is the lowest root of $E[r] + \int_{r^*}^{\bar{r}} \rho_0(r)f(r)dr + F(r^*)L = I - A$, $r^* \in (0, \bar{r})$, $\rho_0(r^*) \leq L$. **(iii)** Short-term debt $d = r^*$: default iff $r < r^*$, creditors liquidate (no renegotiation, $\rho_0(r) < L$). $dr^*/dA = -1/\{f(r^*)[L - \rho_0(r^*)]\} < 0$. **(iv)** $r^* > \hat{r}$: investors get at most $\max\{L/2, \rho_0(r)\}$ in each state, below the old contract on $[0, r^*) \Rightarrow$ cannot break even: no financing. **(v)** $r^* < \hat{r}$: $\mathcal{P}^c(r^*) = I - A - F(r^*)L/2$; financing needs a higher cutoff $r^{**} \in (r^*, \hat{r}]$, or fails. Lenders still break even; the borrower loses $\text{NPV}(r^*) - \text{NPV}(r^{**}) > 0$. Example ($r \sim U[0,1]$, $\rho_0 = 0.12 + 0.2r$, $\rho_1 = 0.6 + r$, $L = 0.5$, $I - A = 0.75$): $r^* = 0.080659$, $r^{**} = 0.3$, loss 0.063681.

14. Two-period sovereign bond (original)

At $t = 1$: $V^c = u(y + B')$, $V^d = u((1-\lambda)y)$, so default iff $b = -B' > \lambda y$ (independent of u). Thresholds $\lambda y_L = 0.08$, $\lambda y_H = 0.12$: $D = \emptyset$ for $B' \geq -0.08$, $\{y_L\}$ for $-0.12 \leq B' < -0.08$, both states below -0.12 (ties repay). $q = (1-\delta)/1.05$: 0.952381 at $0.05, -0.05, -0.08$; 0.761905 at $-0.10, -0.12$; 0 at -0.15. Risky region: $1 + r(B') = 1.3125$, spread 0.2625; lender gets $0.08/0.0761905 = 1.05$. $B^* = -0.12$ raises 0.091429 (safe maximum 0.076190); $B' \in [-0.10, -0.08)$ is dominated by the safe bond $b_s = 0.8b$; beyond B^*, $q = 0$.

15. Price schedule with persistent output (original)

Default iff $b > \kappa(y') = y' - \min\{y', 0.9\} = 0$, 0.1, 0.3. $D(B_2') \subseteq D(B_1')$ for $B_2' \geq B_1'$ ($u(y' + B')$ rises in B', $u(h(y'))$ fixed) $\Rightarrow q$ nondecreasing in B'; D depends on y' only, q on current y. $q(B', y)$ for $y = 0.8, 1.0, 1.2$: $B' \geq 0$: 0.961538; $-0.1 \leq B' < 0$: 0.384615, 0.769231, 0.865385; $-0.3 \leq B' < -0.1$: 0.096154, 0.192308, 0.576923; $B' < -0.3$: 0. Spreads at $B' = -0.05$: 1.56, 0.26, 0.115556 (persistence). I.i.d. $(0.3, 0.4, 0.3)$: $q = 0.673077$ for all y. $B^*(0.8) = B^*(1.0) = -0.1$ (raise 0.038462, 0.076923), $B^*(1.2) = -0.3$ (0.173077): procyclical borrowing limit.

16. Claim 3: no inflows when default is possible (original)

(a) $V^d(y) = u(y) + \frac{\beta}{1-\beta}E[u]$: $V^d(0.9) = -0.150587$, $V^d(1.1) = 0.050084$. (b) $y \in D(B)$ and $V^c \geq u(y + B - q(\hat{B})\hat{B}) + \beta E V^o(\hat{B}, \cdot)$ give $u(y) - u(y + B - q(\hat{B})\hat{B}) > \beta E[V^o(\hat{B}, \cdot) - V^d] \geq 0$,

so $B - q(\hat{B})\hat{B} < 0$. (c) $c = 0.938095$, inflow 0.038095; $V^c \geq V^d(0.9) + 0.041457$: no default. (d) $V^c(0, y) \geq u(y) + \beta E V^d = V^d(y)$, so $D(0) = \emptyset$: borrowing from $B = 0$ is possible. (e) Holds for any $h(y) \leq y$ (indeed $B - q(\hat{B})\hat{B} < -(y - h(y))$) and state by state with Markov output; only $\theta = 0$ matters. (f) With $\theta > 0$ the right side gains $\beta\theta E[V^o(\hat{B}) - V^o(0)] \leq 0$: default as a fresh start can beat a small inflow.

17. Certifying risk-free debt (original)

(a) $V^d(0.9) = -0.352614$, $V^d(1.1) = -0.151943$. (b) Rolling b over at $1/(1+r)$: $c = y - s$, $s = rb/(1+r)$; $V^c(-b, y) \geq W = \ln(y - s) + 9E\ln(\tilde{y} - s)$; $W \geq V^d \Rightarrow D(-b) = \emptyset$. (c) $\partial\Delta/\partial y = s/[y(y - s)] > 0$: test at y_L; with $\lambda = 0$, $\Delta < 0$ for all $s > 0$. (d) $G(s) = 5.5\ln(1 - s/0.9) + 4.5\ln(1 - s/1.1) - 10\ln 0.98$; Newton $0 \to 0.019803 \to 0.019602$; $\bar{s} = 0.019602$, $\bar{b} = 21\bar{s} = 0.411642$. (e) $\bar{b} \leq -\bar{B} = 0.42$ (lower bound); $d\bar{s}/d\lambda = 0.980002$, $d\bar{b}/d\lambda = 20.580043$; capacity $0.01 \to 0.42$ as $\lambda : 0 \to 0.02$, $0.42 \to 0.0375$ as $\theta : 0 \to 0.5$.